

Person-Specific Closed-Loop Neuromodulation in Autism

A Critical Review of the Evidence and a Proposed Computational Framework

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Abstract

Proposals to apply bidirectional brain–computer interfaces (BBCIs) to autism have typically inherited a model of autistic neurobiology that the empirical literature no longer supports: uniform long-range and interhemispheric underconnectivity, to be corrected by driving connectivity toward neurotypical values. This review evaluates that premise against current peer-reviewed evidence and finds it untenable. Connectivity differences in autism are idiosyncratic and heterogeneous rather than uniformly deficient, with both hyper- and hypo-connectivity depending on individual, network, and developmental stage; a substantial portion of the original underconnectivity literature is confounded by head motion; and group-level connectivity effects are small enough that reproducible brain-wide association requires thousands of participants. A second common premise—that neural entropy must be pharmacologically elevated to enable rewiring—is likewise unsupported and, in early development, appears directionally reversed. Two findings nevertheless permit a defensible reformulation. First, densely sampled within-individual functional mapping recovers the reliability that group-level inference lacks, making person-specific modelling tractable. Second, normative modelling provides a principled statistical treatment of heterogeneity in which the target of intervention is each individual’s own deviation profile rather than a population mean. On this basis we propose reframing the problem from normalisation of connectivity toward person-specific closed-loop control, in which neural complexity is an estimated state variable steered toward an individual optimum rather than an injected input, pharmacological entropy induction is replaced by mechanistically interpretable neuromodulation, and outcomes are consented, quality-of-life-relevant endpoints selected with autistic co-investigators. We survey the realistic technical ceiling—timing-dependent cortico-cortical paired associative stimulation, closed-loop stimulation approved only for epilepsy and Parkinson’s disease, and n-of-1 psychiatric neuromodulation—and conclude that no current interface induces durable large-scale interhemispheric connectivity change in humans. We therefore specify a complete in-silico methodology, grounded in connectome-based neural mass modelling, explicit synaptic plasticity rules, physically realistic stimulation field models, network control theory, and preregistered reproducible simulation practice, framed explicitly as hypothesis-generating rather than as a device proposal.

Keywords: autism; brain–computer interface; closed-loop neuromodulation; functional connectivity; normative modelling; neural complexity; network control theory; whole-brain modelling; neuroethics

1 Introduction

The prospect of using a bidirectional brain–computer interface (BBCI) to support autistic people has an obvious intuitive appeal. If autism involves atypical communication between brain

regions, and if an interface can both read neural activity and write to it, then the interface might be used to strengthen the connections that are weak. Proposals along these lines recur in the computational and engineering literature, and they typically share a common architecture: characterise autism as a connectivity deficit, define a target connectivity pattern derived from neurotypical reference data, and drive the system toward that target.

Each step of that architecture is in tension with the current evidence. The characterisation of autism as uniform underconnectivity has been substantially revised. The choice of a neurotypical reference as the target assumes a homogeneity that the data do not show. And the technical capability to induce durable, large-scale connectivity change in humans does not presently exist.

This does not mean the underlying scientific instinct is wrong. Three components of the intuition survive scrutiny and are, in fact, well aligned with the frontier: that closed-loop coupling of decoding and stimulation is the right control architecture; that the plasticity state of the system modulates the efficacy of any intervention; and that the brain is usefully treated as a controllable dynamical system. What must change is the target, the model, and the framing.

This paper has two objectives. The first is a critical review of the peer-reviewed literature bearing on each premise of the BBCI-for-autism proposal: connectivity, complexity and entropy, plasticity induction, neuromodulation and interface capability, computational modelling, and research ethics. Where the literature contradicts a common premise, we state so directly and identify the defensible reformulation. The second objective is constructive: to specify a complete in-silico methodology for a study that could be defended at doctoral level, built on the reformulated premises. Throughout, we distinguish well-established findings from contested or preliminary ones, and report effect sizes and sample sizes where available.

2 Scope and approach

This is a critical narrative review rather than a systematic review, and it makes no claim to exhaustive coverage. Sources were restricted to peer-reviewed publications identified through academic literature search across neuroscience, neuroimaging, neuroengineering, computational modelling, and autism research. Preprints are cited only where explicitly labelled as such and are not load-bearing for any conclusion. Where a claim rests on a single small study, or on evidence with recognised risk of bias, this is stated in the text. Section 13 summarises the evidence quality across the domains reviewed.

Terminology follows the preferences most commonly expressed by autistic self-advocates and increasingly adopted in the field: identity-first language (“autistic person”), and avoidance of “cure” and “normalisation” framing except where describing positions held in the literature under review.

3 Connectivity in autism: from uniform underconnectivity to idiosyncrasy

3.1 The historical underconnectivity account

The influential formulation of autism as a disorder of cortical underconnectivity derives from Just and colleagues, who reported reduced functional synchronisation between frontal and posterior language regions during sentence comprehension in autistic adults [1]. The account was extended across task domains and became, for roughly a decade, the dominant systems-level

model of autistic neurobiology [2]. Its appeal was considerable: it offered a single mechanistic principle—insufficient long-range integration—capable of explaining difficulties spanning language, social cognition, and executive function.

3.2 Heterogeneity and idiosyncrasy

That unified account has not survived. In a study that reframed the field, Hahamy, Behrmann and Malach analysed resting-state functional connectivity in autistic adults and found regions of both increased and decreased connectivity relative to controls [3]. Their central inference was that the group difference is not a consistent direction of deviation but a consistent *loss of conformity*: autistic connectivity patterns are idiosyncratically distorted relative to the canonical template, and the magnitude of an individual's distortion in homotopic interhemispheric connectivity correlated with behavioural symptom severity. The deficit, on this reading, is in the typicality of the pattern rather than in its overall strength.

This has been repeatedly corroborated across independent samples and analytic approaches. Nunes and colleagues, using ABIDE I and II ($n = 422$ autistic, $n = 424$ comparison), found idiosyncratic organisation of intrinsic connectivity networks that correlated with diagnostic observation scores [4]. Benkarim and colleagues ($n = 157$ autistic, $n = 172$ comparison) found increased idiosyncrasy concentrated in default-mode, somatomotor and attention networks, and demonstrated that this idiosyncrasy is itself a driver of the inconsistency observed across conventional case–control studies [5]. Buch and colleagues identified latent connectivity dimensions and reproducible subgroups with distinct molecular associations [6]. Uddin and colleagues had earlier established that the direction of connectivity difference depends systematically on developmental stage, with hyperconnectivity more characteristic of childhood and hypoconnectivity of adolescence and adulthood [7, 8].

The largest well-powered multi-cohort test to date, by Holiga and colleagues across four independent cohorts including EU-AIMS LEAP, found alterations that were statistically reproducible but modest and heterogeneous—hypoconnectivity in sensory-motor regions alongside hyperconnectivity in prefrontal and parietal hubs [9]. The authors' own conclusion is the operative one for any intervention proposal: the moderate association with clinical symptoms, together with substantial overlap between autistic and comparison distributions, limits the present usability of these alterations.

3.3 Normative modelling as the principled treatment of heterogeneity

If deviation is person-specific, then group-mean contrasts are the wrong statistical instrument. Marquand and colleagues proposed reconceptualising psychiatric conditions as deviations from a normative reference range, analogous to paediatric growth charts, with inference conducted at the level of the individual rather than the group [10]. Applied to autism, this approach is revealing. Zabihi and colleagues, working in EU-AIMS LEAP data, found that some autistic individuals show widespread positive deviations from population norms in cortical anatomy while others show widespread negative deviations, with few consistent group-level differences [11, 12]. Bethlehem and colleagues reached a convergent conclusion for cortical thickness [13]. The implication is direct: the “average autistic brain” is a statistical artefact of averaging over individuals whose deviations point in different directions.

This is the formal foundation for a person-specific approach. If the deviation profile is individual, any fixed normalisation target is wrong by construction, and the appropriate target is

each individual's own deviation from their own reference—a conclusion the normative modelling literature reaches independently on statistical grounds.

3.4 The head-motion confound

A methodological problem compounds the interpretive one. Power and colleagues, Van Dijk and colleagues, and Satterthwaite and colleagues independently demonstrated that in-scanner head motion produces spatially structured spurious functional connectivity, systematically inflating short-range correlations and deflating long-range ones [14, 15, 16]. This is precisely the spatial signature reported as autistic underconnectivity, in a population that moves more in the scanner. Motion-controlled reanalyses concluded that a substantial portion of the early findings may be artefactual [17]. Motion remains a first-order confound for any study comparing clinical and comparison groups, and any connectivity claim that does not report and control framewise displacement should be treated as provisional.

3.5 Interhemispheric connectivity and lateralisation

Findings specific to interhemispheric communication exist but are modest. Anderson and colleagues reported decreased interhemispheric functional connectivity [18]; meta-analytic and longitudinal work has reported differences in corpus callosum volume and microstructure [19, 20]; and Dinse and colleagues found disrupted interhemispheric synchrony during natural sleep in autistic toddlers, a design that elegantly circumvents both task-compliance and much of the motion problem [21]. Separately, a literature documents atypical or reduced hemispheric lateralisation, particularly for language [22, 23].

Taken together, these support the existence of interhemispheric differences in some autistic individuals at some developmental stages. They do not support “hemispheric isolation” as a general characterisation of autism. That framing overstates a modest, heterogeneous, inconsistently replicated finding, partly confounded by motion, and superseded in its uniform-deficit form by the idiosyncrasy literature.

3.6 Reliability and effect size

The final constraint is statistical. Marek and colleagues, analysing samples totalling approximately 50,000 individuals, showed that brain-wide association studies linking individual differences in functional connectivity to phenotype require thousands of participants for reproducibility, and reported a median reproducible effect size on the order of $r \approx 0.01$ [24]. Given that the median neuroimaging study enrolls roughly 25 participants, most published brain-behaviour associations in this literature are underpowered by orders of magnitude. A peer-reviewed rejoinder argues that multivariate methods can replicate at moderate sample sizes [25], and the debate is live; the conservative reading is that univariate group-level connectivity biomarkers are not currently a sound basis for intervention design. Systematic review of test-retest reliability supports the same caution [26].

The counterpoint is decisive for the reformulation proposed here. Precision functional mapping—dense repeated sampling within individuals—recovers the reliability that group inference lacks. Gordon and colleagues, acquiring approximately five hours of resting-state data per participant in ten adults, found that individual network estimates stabilise with sufficient data (on the order of tens of minutes minimum, with substantially more required for fine detail), and that individual maps contain reliable features obscured by group averaging

[27]. Converging work by Laumann, Gratton, Poldrack and colleagues supports the same conclusion [28, 29, 30]. Group-level inference is unreliable at typical sample sizes; individual-level inference with adequate per-person data is not. This asymmetry is the enabling condition for person-specific modelling.

4 Neural signal complexity and entropy in autism

4.1 Empirical findings

Proposals that invoke entropy as a lever for plasticity generally assume that autistic neural dynamics are insufficiently entropic and require elevation. The developmental evidence points, where it points at all, in the opposite direction.

Bosl and colleagues, examining resting electroencephalography in infants at elevated familial likelihood of autism ($n = 79$), found *reduced* modified multiscale entropy, most pronounced frontally, with classification accuracy exceeding 80% at nine months [31]; subsequent work extended the approach [32]. A peer-reviewed critique cautioned against conflating population-level and individual-level risk inference from these data, and should be read alongside the original [33]. Catarino and colleagues reported reduced multiscale entropy at coarser timescales in autistic adults, predominantly over posterior regions [34]. Ghanbari and colleagues reported complexity differences in magnetoencephalography [35], and Takahashi reviewed altered complexity across developmental conditions [36].

The honest summary is that findings are measure-, region-, task- and age-dependent, and no single direction holds universally. But where a direction emerges in early development, it is toward reduced rather than elevated complexity. A framework premised on the need to inject entropy is therefore not merely mechanistically underspecified—it may be directionally wrong.

Measurement caveats are substantial and frequently neglected. Lempel–Ziv complexity, approximate entropy, sample entropy, multiscale entropy and permutation entropy are not interchangeable; they are sensitive to series length, stationarity, sampling rate, and embedding parameters, and can diverge on the same data. Any claim about “entropy” must specify the estimator.

4.2 Criticality and the inverted-U hypothesis

A more defensible formulation is available. The criticality literature holds that cortical dynamics operate near a critical point at which complexity, dynamic range, and information-processing capacity are jointly maximised [37, 38]. On this account the relevant quantity is not entropy per se but *distance from criticality*, with deviation in either direction being costly. O’Byrne and Jerbi provide a critical appraisal of how far the evidence supports strong criticality claims [39], and Toker and colleagues report that waking cortical dynamics sit slightly on the chaotic side of the critical boundary, with loss of consciousness corresponding to movement away from it [40].

This framing makes an entropy-based target empirically tractable for the first time: rather than injecting noise, a controller estimates the individual’s position relative to an optimum and steers toward it, with the inverted-U relationship itself constituting a falsifiable prediction. The optimum has not been established for autism specifically, and criticality metrics remain contested; this is a hypothesis to be tested, not an established mechanism.

5 Entropic brain theory and the induction of plasticity

5.1 The entropic brain hypothesis

Carhart-Harris and colleagues proposed that, within upper and lower bounds beyond which conscious function degrades, the entropy of spontaneous brain activity indexes the richness of conscious state, and that psychedelics act by elevating it [41, 42]. The REBUS formulation situates this within a predictive-processing account in which high-level priors are relaxed [43]. Empirically, the construct is operationalised largely through signal diversity measures, with reports of increased spontaneous magnetoencephalographic diversity under ketamine, LSD and psilocybin [44, 45].

The theory's falsifiability is limited by its measurement. Reported psychedelic entropy effects are inconsistent across estimators, with some measures showing robust effects and others null results on the same data—empirical evidence that these metrics do not index a single construct. Head motion is also a serious confound in psychedelic neuroimaging. Entropic brain theory is a productive research programme, not a settled mechanism.

5.2 Receptor-informed whole-brain modelling

The most methodologically useful contribution of this literature is not the theory but the modelling approach it generated. Deco and colleagues constructed a dynamic mean-field whole-brain model in which serotonin 5-HT_{2A} receptor density, derived from positron emission tomography, modulates local neuronal gain, and showed that this reproduces the functional connectivity dynamics observed under LSD [46]. Preller and colleagues attributed LSD-induced connectivity changes specifically to the 5-HT_{2A} receptor [47], and Singleton and colleagues linked receptor-informed modelling to a flattening of the brain's control energy landscape via network control theory [48].

The methodological lesson generalises well beyond psychedelics: the interpretable way to manipulate a global dynamical regime *in silico* is spatially structured gain or excitation–inhibition modulation informed by empirical receptor or physiological maps. Adding unstructured Gaussian noise is not a model of anything.

5.3 Psychoplastogens and critical-period reopening

Separately, there is now strong evidence that serotonergic psychedelics promote structural neuroplasticity. Ly and colleagues demonstrated increased dendritic arbor complexity and spine density [49]; Shao and colleagues showed rapid and persistent dendritic spine growth in frontal cortex *in vivo* [50]; Vargas and colleagues identified intracellular 5-HT_{2A} receptors as necessary for the plasticity-promoting effect [51]; and Moliner and colleagues showed direct binding to the BDNF receptor TrkB [52].

The most conceptually relevant finding is critical-period reopening. Nardou and colleagues showed that MDMA reopens the critical period for social reward learning in mice via oxytocin-dependent metaplasticity [53], and subsequently that this is a property shared across ketamine, LSD, MDMA, psilocybin and ibogaine, with the duration of reopening tracking each compound's subjective duration in humans [54]. This is a genuine mechanistic bridge between pharmacology and developmental plasticity. Its limits must be stated with equal clarity: it is rodent work, on social reward learning specifically, with no human translation, and a reopened social-reward-learning window is not a model of autism.

5.4 Psychedelics in autistic populations

Direct evidence in autistic people is limited to one modern trial. Danforth and colleagues conducted a randomised, double-blind, placebo-controlled pilot of MDMA-assisted therapy for social anxiety in autistic adults [55]. With $n = 12$ (8 MDMA, 4 placebo) and two eight-hour sessions, improvement on the Liebowitz Social Anxiety Scale was significantly greater in the MDMA arm ($P = 0.037$), with a large placebo-subtracted effect size ($d = 1.4$) at the primary endpoint and $d = 1.1$ at six months. The result is encouraging but severely underpowered—the confidence intervals cross or approach zero—and the trial was sponsor-funded. It is a signal warranting replication, not an evidence base.

The historical record is a necessary counterweight. In the 1960s, LSD was administered to autistic and institutionalised children in a series of studies characterised by absent consent, inadequate methodology, and documented harm [56, 57, 58]. Any contemporary proposal involving psychoactive intervention in a population that is frequently minor and frequently communication-diverse must engage this precedent explicitly rather than treat it as historical curiosity.

Finally, pharmacodynamics are not uniform in this population. Meta-analysis by Gabriele and colleagues found elevated blood serotonin in approximately 28% of autistic individuals in whole blood and 23% in platelet-rich plasma [59], with implications for both response and risk in any serotonergic intervention.

5.5 Assessment

On balance, the pharmacological component should be removed from the core of a BBCI research programme. The concept it is meant to serve—that intervention efficacy depends on the plasticity state of the target system—can be dissociated from pharmacology and pursued through neuromodulation, at a fraction of the ethical and regulatory burden. Receptor-informed whole-brain modelling should be retained as an in-silico source of a mechanistically interpretable gain-modulation operator, labelled explicitly as a modelling analogy rather than a treatment proposal.

6 Neuromodulation and brain–computer interfaces in autism

6.1 Non-invasive brain stimulation

Repetitive transcranial magnetic stimulation has been applied to autism for over a decade, most extensively by Casanova and colleagues using low-frequency protocols over dorsolateral prefrontal cortex [60], with a double-blind randomised trial of deep rTMS reported by Enticott and colleagues [61]. The meta-analysis by Barahona-Corrêa and colleagues synthesised 23 reports—four case reports, seven uncontrolled studies, and twelve controlled studies—and found statistically significant but moderate effects on repetitive and stereotyped behaviour, social behaviour, and executive-function error rates [62]. The same analysis found that most included studies carried moderate-to-high risk of bias, principally through absent participant or evaluator blinding, with evidence of publication bias, and that only five studies reported durability at approximately six months. Subsequent systematic review reached similarly cautious conclusions [63], and more recent syntheses restricted to randomised controlled trials have been more conservative still, attributing part of the earlier positive signal to inclusion of uncontrolled designs.

Transcranial direct and alternating current stimulation have a smaller and similarly heterogeneous evidence base. Transcranial focused ultrasound is a genuinely promising emerging modality for precise and deep targeting, with human and non-human primate demonstrations of focal neuromodulation [64, 65], but it is early-stage with respect to safety, dosimetry and efficacy, and untested for durable connectivity change in this population.

The overall verdict is that no non-invasive stimulation protocol has established efficacy for autism. This is not a reason to abandon the modality; it is a reason not to treat any existing protocol as a validated actuator.

6.2 Neurofeedback

Neurofeedback, including mu-rhythm protocols motivated by mirror-neuron accounts [66], has been repeatedly undermined by blinding and placebo problems. Thibault and Raz argue that clinical benefit in neurofeedback is substantially attributable to placebo mechanisms even where present [67], and systematic review supports caution [68]. Studies lacking adequate sham control should not be treated as evidence of specific efficacy.

6.3 Brain–computer interfaces in autism

A small but directly relevant applied literature exists and is often overlooked in engineering-led proposals. Amaral and colleagues conducted a single-arm feasibility trial of a P300-based electroencephalographic BCI combined with immersive virtual reality to train joint attention in fifteen autistic adults over seven sessions [69]; the associated BCIAUT-P300 benchmark dataset has since supported methodological work across the field [70]. Mercado and colleagues developed and evaluated BCI-based neurofeedback games for autistic children [71].

These are feasibility and engineering studies rather than efficacy trials, and should not be cited as evidence of clinical benefit. They do, however, establish that closed-loop electroencephalographic interfaces are a live, publishable, and ethically navigable research area in this population—and they constitute the natural empirical partner for a simulation programme.

7 Bidirectional interfaces and the induction of connectivity

7.1 Taxonomy

“Bidirectional” is used loosely. It is useful to distinguish read-only interfaces that decode neural activity; write-capable interfaces that deliver open-loop stimulation; and genuinely closed-loop systems in which decoded state determines stimulation in real time. Only the third constitutes bidirectional control in the sense relevant here.

7.2 Closed-loop systems in clinical use

The clearest demonstration of true bidirectional operation is somatosensory feedback in motor prostheses: Flesher and colleagues showed that intracortical microstimulation of human somatosensory cortex evokes graded percepts [72] and that closing the loop with tactile feedback substantially improves robotic arm control [73].

In clinical neuromodulation, responsive stimulation for drug-resistant focal epilepsy has been available since 2013 following pivotal trial evidence [74]. Adaptive deep brain stimulation, in which stimulation is titrated to a sensed physiological biomarker, was demonstrated in principle by Little and colleagues [75] and received United States regulatory approval in

February 2025 for Parkinson's disease on the basis of the ADAPT-PD trial [76]. It is important to be precise about scope: these approvals cover epilepsy and Parkinson's disease. No closed-loop system is approved for any psychiatric or neurodevelopmental indication.

In psychiatry the work remains at proof-of-concept scale. Scangos and colleagues delivered biomarker-triggered closed-loop stimulation in a single individual with treatment-resistant depression, using amygdala gamma activity to gate ventral capsule stimulation, and reported rapid and sustained improvement while explicitly cautioning that generalisation remains to be established [77]. Alagapan and colleagues subsequently demonstrated that cingulate dynamics track recovery under deep brain stimulation [78]. Closed-loop memory stimulation has been shown to improve encoding in epilepsy patients with implanted electrodes [79].

7.3 Paired stimulation and induced connectivity

The closest empirical analogue to the aspiration of “strengthening the connection between two regions” is paired associative stimulation. The principle derives from spike-timing-dependent plasticity: synaptic strength is modified as a function of the relative timing of pre- and postsynaptic activity, with millisecond-scale sensitivity and sign reversal across the timing window [80, 81]. Stefan and colleagues translated this into a human protocol pairing peripheral and cortical stimulation [82].

Cortico-cortical paired associative stimulation (ccPAS) extends this to pairs of cortical sites. Koch and colleagues demonstrated Hebbian and anti-Hebbian modulation of human cortico-cortical connections as a function of stimulation order and interval [83], and subsequent work has established effects across premotor–motor and visual pathways [84, 85]. Of particular relevance, Rizzo and colleagues showed that paired stimulation of left and right motor cortex shapes interhemispheric inhibition according to a Hebbian timing rule [86]—the literal empirical instantiation of interhemispheric connectivity modulation, achieved with interstimulus intervals on the order of eight milliseconds, and abolished at non-physiological or randomised intervals.

Complementary evidence comes from activity-dependent stimulation in primates. Jackson, Mavoori and Fetz showed that an autonomous implant delivering stimulation contingent on recorded spikes induces lasting reorganisation of motor cortical output [87], and Seeman and colleagues extended paired-stimulation plasticity in sensorimotor cortex [88]. Paired vagus nerve stimulation provides a further route to targeted plasticity [89].

7.4 The realistic ceiling

Effect sizes in ccPAS are modest, durability typically ranges from minutes to hours with occasional reports of longer persistence, and results are sensitive to protocol parameters and individual state. Synthesising across this literature: no current bidirectional interface induces durable, large-scale interhemispheric connectivity change in humans. What is achievable is timing-dependent modulation of specific cortico-cortical pathways, closed-loop stimulation in two approved neurological indications, and single-case psychiatric neuromodulation.

This gap between aspiration and capability is the principal argument for pursuing the question computationally first. A simulation study is not a weaker version of a device study; given the present state of the field, it is the appropriate instrument.

8 Computational models of autism

8.1 Excitation–inhibition balance

Rubenstein and Merzenich proposed that autism involves an elevated ratio of excitation to inhibition in key circuits [90], a framework elaborated by Nelson and Valakh [91]. Its causal status is contested. Antoine and colleagues found elevated excitation–inhibition ratio across four genetic mouse models yet normal layer 2/3 firing rates, arguing that the change is homeostatic and compensatory rather than pathogenic [92]. Sohal and Rubenstein offer a nuanced synthesis [93]. Nonetheless, optogenetic correction of prefrontal excitation–inhibition balance rescues social behaviour in a genetic mouse model [94], establishing that the ratio is at minimum a manipulable lever with behavioural consequences. For modelling purposes it is best treated as a candidate control variable rather than settled aetiology.

8.2 Predictive processing accounts

A family of Bayesian accounts proposes that autistic perception reflects atypical weighting of prior expectation against sensory evidence: attenuated priors [95], inflexibly high estimation of prediction error [96], aberrant precision [97], or impaired prediction more generally [98]. Empirical tests are mixed and the accounts are not mutually exclusive. They provide a principled bridge between circuit-level parameters (gain, precision) and phenomenology, which is valuable for a modelling programme.

8.3 Variability is not complexity

Dinstein and colleagues reported increased trial-to-trial variability of evoked cortical responses in autism [99]. This is sometimes conflated with the complexity findings reviewed in Section 4, but variability, noise, and entropy are mathematically distinct quantities that can move independently: a system can be more variable and less complex simultaneously. Any model that treats these as interchangeable is underspecified.

8.4 Whole-brain generative modelling

Connectome-based generative modelling provides the mature computational infrastructure for this problem. The Virtual Brain framework couples neural mass models through empirically derived structural connectivity to generate simulated neuroimaging signals [100], and Deco and Kringelbach set out the case for whole-brain computational connectomics in neuropsychiatric research, including in-silico perturbation [101]. The *neurolib* framework provides an accessible implementation of excitatory–inhibitory mean-field models with diffusion-derived structural connectivity [102]. Underlying population models—Wilson–Cowan [103] and Jansen–Rit [104]—are well characterised. Dynamic causal modelling provides a complementary approach for inferring directed effective connectivity [105].

Direct application of whole-brain generative models to autism data remains sparse. This is a genuine gap and an opportunity.

8.5 Network control theory

The rigorous formalisation of “guiding neural activity” is control theory applied to brain networks. Gu and colleagues characterised the controllability properties of structural brain networks [106]; Muldoon and colleagues extended this to stimulation-based control in nonlinear

network models [107]; and Stiso and colleagues showed that white-matter architecture predicts the effects of direct electrical stimulation on state transitions [108].

The framework has been substantively criticised. Tu and colleagues challenged claims about single-node controllability and questioned the macroscopic linearity assumption [109]; Pasqualetti and colleagues responded in defence [110]. The defensible position is that linear controllability analysis is a useful and tractable approximation whose assumptions must be stated and whose conclusions must be checked against nonlinear simulation.

9 Data resources

ABIDE I and II provide the largest openly available autism neuroimaging aggregation, comprising over two thousand datasets across multiple sites [111, 112]. Their principal limitations for the present purpose are multi-site heterogeneity, motion, and short per-participant acquisition durations that preclude precision-mapping-style individual characterisation. EU-AIMS LEAP offers deep, longitudinal, harmonised phenotyping and is the strongest resource for heterogeneity and normative work [113]. The Healthy Brain Network provides a large transdiagnostic paediatric resource [114]. Standardised preprocessing [115] and site harmonisation [116] are prerequisites rather than options.

For a person-specific programme, these resources support the estimation of population priors and provide validation targets, but do not by themselves support full individual identifiability. Claims at the individual level must be bounded accordingly, or supported by densely sampled acquisitions.

10 Ethical framing

Ethical considerations here are not a compliance appendix; they determine whether the research question is well posed.

The neurodiversity paradigm reframes autistic social difficulty as a bidirectional mismatch between differently constituted people rather than a unilateral deficit—Milton’s double empathy problem [117], for which supporting evidence has accumulated substantially. Kapp and colleagues documented the relationship between neurodiversity framing and identity [118]. Botha and Cage document researchers’ constructions of autistic people and the resulting tensions in the field [119]. Pellicano and den Houting argue for a shift from deficit-focused “normal science” toward neurodiversity-informed research [120], and participatory standards for autism research are now well specified [121, 122].

There is also a documented mismatch between research funding and community priorities: autistic people and families consistently prioritise services, mental health, and quality of life over aetiological and biological research [123]. A project framed as connectivity normalisation sits on the wrong side of that mismatch. A project framed around consented, self-identified, quality-of-life-relevant outcomes—sensory-overload distress, co-occurring anxiety, sleep, interoception, executive-function support—sits on the right side, and has the additional methodological advantage of measurable endpoints that do not conflate benefit with reduction in observable autistic characteristics. Outcome measures that index “autisticness” rather than wellbeing are poorly suited to this purpose.

Intervening in a developing brain raises further issues: consent and assent for minors, the distinction between treatment, enhancement and normalisation, and questions of identity and

authenticity. These are increasingly reflected in policy. Ethical priorities for neurotechnology have been articulated in the scientific literature [124]; cognitive liberty has been developed as a legal framework [125]; several jurisdictions have enacted neural data protections; and UNESCO adopted a global recommendation on the ethics of neurotechnology in 2025 [126].

The practical consequence for the present programme is that the research question should be posed as: *can a person-specific closed-loop system steer an individual's neural dynamics toward a state associated with an outcome that individual has chosen?*—not *can connectivity be normalised?*

11 Common pitfalls in simplified BBCI simulations

Simplified simulations of BBCI operation are a reasonable pedagogical starting point, but a recurring set of failure modes renders their conclusions uninformative. They are worth stating explicitly, since they are easy to reproduce inadvertently and difficult to detect from results alone.

1. **Random processes standing in for neural dynamics.** A matrix evolved by random perturbation has no oscillatory structure, no excitation–inhibition architecture, and no mechanistic content. It cannot support inference about connectivity because it does not model connectivity.
2. **Noise injection described as entropy.** Adding scaled random values is neither entropy in the information-theoretic sense nor in the sense used by entropic brain theory. If no entropy is computed, no entropy claim can be made.
3. **Classification without ground truth.** A classifier trained on synthetic activity with synthetically assigned labels will achieve high accuracy by construction. This demonstrates nothing about decoding intent; it demonstrates that the labels were recoverable from the data that generated them.
4. **Assuming the conclusion in the actuator.** Implementing “synchronisation” by directly moving one region’s values toward another’s is definitional rather than derived. The interesting question is whether a plausible plasticity mechanism produces convergence, not whether convergence can be imposed.
5. **Absent plasticity rules.** Where plasticity is the central claim, an explicit learning rule with cited parameterisation is mandatory. Its absence means the model contains no plasticity.
6. **No empirical grounding.** Without connectomes, physiological parameters, or validation targets drawn from data, a model has no purchase on the phenomenon it names.
7. **Absent inference.** Simulation results require surrogate and null models, sensitivity analysis, and uncertainty quantification. Without these, a result is an anecdote about one parameter setting.
8. **Insufficient scale.** Connectivity claims require enough units and enough time to make connectivity a meaningful quantity. Two regions of ten units over one hundred timesteps does not qualify.
9. **Unmodelled stimulation.** Stimulation implemented as arithmetic addition to selected units

has no dose, no field, and no spatial spread, and therefore no correspondence to any physical intervention.

10. **Unfalsifiable results.** If every reported finding follows deductively from the construction of the code, the study has no empirical content. This is the diagnostic question to ask of any simulation: what result would have counted as failure?

The corrective is not additional complexity for its own sake. It is that each element of the model must be independently motivated, empirically grounded where possible, and capable of producing a result its author did not build in.

12 Proposed methods: an in-silico framework

What follows specifies a computational study designed to be defensible at doctoral level. It involves no human participants, is explicitly hypothesis-generating, and is a precursor to—not a substitute for—empirical work.

12.1 Research question and hypotheses

Question. In individual-level whole-brain generative models fitted to autistic and comparison connectomes, can a person-specific closed-loop controller that estimates an individual's neural complexity and effective connectivity steer the model toward that individual's pre-specified target regime using physically realistic, timing-dependent stimulation?

Hypotheses.

- H1** Timing-dependent paired stimulation increases targeted effective connectivity relative to sham and to reversed-timing control, with the effect present only at interstimulus intervals consistent with spike-timing-dependent plasticity windows.
- H2** Simulated signal complexity follows an inverted-U function of the controlled gain or excitation–inhibition parameter, with a person-specific optimum.
- H3** Steering complexity toward the individual optimum improves the surrogate outcome relative to complexity-agnostic control.

Disconfirmation criteria, pre-specified. If connectivity change under timing-dependent stimulation is statistically indistinguishable from random-timing stimulation, the spike-timing-dependent mechanism is disconfirmed. If H3 fails, the entropy-as-target premise is disconfirmed. Both criteria are registered in advance and reported regardless of outcome.

12.2 Model selection

The primary model is connectome-based whole-brain neural mass modelling—The Virtual Brain [100] or neurolib [102]—using Wilson–Cowan [103] or Jansen–Rit [104] population dynamics with explicit excitatory and inhibitory subpopulations across 68–200 parcellated regions. This level of abstraction is selected because it is empirically constrained by diffusion-derived structural connectivity, generates observables comparable to functional magnetic resonance imaging and electroencephalography, and supports in-silico stimulation and receptor-informed gain modulation. A secondary spiking-network implementation [127, 128] at reduced scale provides mechanistic detail on plasticity dynamics that mean-field models cannot resolve. The trade-off—mean-field tractability and empirical comparability against spiking

biophysical realism—is addressed by running both and reporting convergence or divergence between levels.

12.3 Empirical grounding and individual fitting

Structural connectomes and resting-state functional connectivity are derived from ABIDE I and II [111, 112], with EU-AIMS LEAP [113] and the Healthy Brain Network [114] as replication and normative resources. Preprocessing follows standardised pipelines [115] with explicit motion control—framewise displacement thresholding, scrubbing, and confound regression—given the confounds documented in Section 3.4, and site harmonisation via empirical Bayes methods [116].

Because the thesis is person-specific, model parameters are fitted at the individual level, with formal identifiability and parameter-recovery analysis: parameters are drawn from known distributions, data simulated, models refitted, and recovery accuracy quantified. Bayesian and simulation-based inference methods are used for posterior estimation [129, 130]. Where identifiability is insufficient given available data duration, this is reported as a bound on the claims rather than concealed.

12.4 Synaptic plasticity

Plasticity is implemented explicitly rather than assumed. At the spiking level, a spike-timing-dependent plasticity rule with cited parameterisation [80, 81] governs synaptic modification. At the mean-field level, a rate-based Hebbian rule modifies inter-regional coupling. Both are paired with homeostatic mechanisms—synaptic scaling [131] and inhibitory plasticity clamping excitatory firing to a target rate [132]—to prevent runaway potentiation, a standard and necessary safeguard in plastic network simulation.

12.5 Complexity as a measured quantity

Complexity is computed from simulated signals rather than injected. Estimators include Lempel–Ziv complexity [133], multiscale entropy [134], permutation entropy [135], and state-repertoire or dynamic functional connectivity diversity measures. Because these estimators are not interchangeable, all are reported, with sensitivity to series length, sampling rate and embedding parameters characterised explicitly. Online estimation feasibility and latency are assessed for closed-loop use.

Manipulation is mechanistic: excitation–inhibition ratio, or spatially structured gain modulation informed by receptor density maps following the approach of Deco and colleagues [46, 47]. Unstructured noise injection is not used.

12.6 The interface as a control system

The closed loop is formalised with explicit components: a state estimator (Kalman or particle filtering, or Bayesian estimation of latent state and complexity); a control policy, with linear-quadratic regulation and model predictive control as baselines and reinforcement learning as an extension; and an actuation model. Sampling rate, loop latency, and stability are analysed explicitly, including Lyapunov stability where tractable. Controllability analysis follows established network control methods [106, 107, 108], with the linearity assumption stated and its limitations acknowledged [109, 110]; conclusions drawn from linear analysis are checked against nonlinear simulation.

12.7 Physically realistic stimulation

Stimulation is modelled as a physical field rather than an arithmetic operation, using established field modelling tools for transcranial magnetic and electrical stimulation [136, 137], with realistic dosing and spatial spread. For paired protocols, interstimulus intervals follow the empirical ccPAS literature [83, 86] and spike-timing-dependent plasticity windows.

12.8 Experimental design

Conditions comprise sham stimulation; random-timing stimulation; reversed-timing (anti-Hebbian) control; complexity manipulation alone; stimulation alone; and combined intervention. The design is completed by parameter sweeps across the physiologically plausible range, sensitivity analysis, and null models—phase-randomised surrogates [138] and degree-preserving network nulls [139]—to establish that observed effects exceed what network topology and spectral content alone produce.

12.9 Outcomes and inference

Primary and secondary outcomes are pre-specified. Because sample size in simulation is arbitrary, null-hypothesis significance testing is close to meaningless: any effect can be made significant by running more seeds. Reporting therefore emphasises effect sizes with confidence intervals, robustness maps across parameter space, Bayesian model comparison, and parameter-recovery results, with multiple-comparison correction applied across the swept parameter space. This point is stated explicitly in the manuscript rather than left implicit, as it is a common source of overclaiming in computational work.

12.10 Validation

Validation proceeds at three levels: face validity (does the fitted model reproduce known dynamical signatures?); construct validity (do simulated group differences reproduce the empirical connectivity [3, 9] and complexity [31, 34] literature?); and predictive validity (does the model predict held-out individual data?). Falsification criteria are those specified in Section 12.1.

12.11 Reproducibility

The study is preregistered [140]. Implementation uses version control, containerisation, pinned dependencies, and controlled random seeds. Code and derived data are deposited openly with a persistent identifier, following FAIR principles [141] and established guidance for reproducible computational research [142].

12.12 Implementation

Python is recommended as the primary environment. The determining consideration is ecosystem maturity: the whole-brain modelling, neuroimaging preprocessing, electrophysiology, inference and machine learning stacks required here are all mature and interoperable in Python, and the collaborator base in computational neuroscience is correspondingly larger. Julia offers genuine advantages in differential-equation performance and has a developing neural-mass ecosystem, and a defensible hybrid architecture uses Python for the modelling, inference and neuroimaging pipeline while delegating performance-critical numerical integration to Julia

through a reproducible interface. Building the entire programme in Julia alone would trade ecosystem depth for performance in a domain where the former dominates project risk.

12.13 Limitations and ethical statement

A simulation cannot establish clinical efficacy or safety, and no such claim is made. Individual-level identifiability is bounded by available data duration and quality. Model results are conditional on the chosen level of abstraction and may not transfer across scales. Any downstream empirical programme arising from this work must be participatory, with autistic co-investigators involved in question formulation and outcome selection [121]; must target consented outcomes rather than reduction in observable autistic characteristics; must abandon normalisation framing; and must comply with emerging neural data protection requirements [126].

13 Evidence quality

Table 1 summarises the strength of evidence across the domains reviewed. The distinction matters because proposals in this area frequently draw on all of these literatures without distinguishing between them.

Table 1: Evidence quality by domain.

Domain	Strength	Notes
Connectivity heterogeneity and idiosyncrasy	Well established	Replicated across independent samples and analytic approaches
Normative modelling of heterogeneity	Well established	Statistically principled; converging results across modalities
Head motion as confound	Well established	Independently demonstrated; affects the historical literature
Individual-level reliability with dense sampling	Well established	Enabling condition for person-specific modelling
Paired associative stimulation of connectivity	Established but modest	Effects small, often transient, protocol-sensitive
Receptor-informed gain modulation in modelling	Established method	Principled alternative to noise injection
Excitation–inhibition imbalance	Contested	Causal versus compensatory status unresolved
Criticality and inverted-U complexity	Contested	Not established for autism specifically
Entropic brain theory	Contested	Measurement fragmented across estimators
Predictive processing accounts	Framework-level	Mixed empirical support; not mutually exclusive
Neuromodulation efficacy in autism	Weak	High risk of bias; blinding inadequate; publication bias
MDMA-assisted therapy in autistic adults	Preliminary	Single pilot, $n = 12$, sponsor-funded
BCI interventions in autism	Feasibility only	Not efficacy trials
Critical-period reopening	No human trans-lation	Rodent, social reward learning specifically

14 Conclusion

The proposal to apply bidirectional brain–computer interfaces to autism, in its common form, rests on premises that the current literature does not support. Autism is not characterised by uniform interhemispheric underconnectivity; connectivity deviations are idiosyncratic, and a substantial portion of the historical evidence for the uniform account is confounded by head motion. Group-level connectivity effects are small enough to require samples orders of magnitude larger than those typically used. Neural complexity in autism is not straightforwardly reduced in a manner that entropy injection would remedy, and in early development the direction of the reported effect runs opposite to the assumption. And no existing interface can induce durable large-scale connectivity change in humans.

None of this makes the underlying research direction unviable. It makes the original formulation unviable, while pointing clearly toward a stronger one. Two findings do most of the work. Normative modelling establishes that the appropriate target is an individual's own deviation profile rather than a population mean. Precision functional mapping establishes that individual-level characterisation is reliable where group-level inference is not. Together they support reframing the problem from normalisation toward person-specific closed-loop control—with complexity as an estimated state variable rather than an injected input, mechanistically interpretable neuromodulation in place of pharmacological entropy induction, and consented quality-of-life outcomes selected with autistic co-investigators in place of connectivity normalisation.

Formulated this way, the programme is scientifically defensible, ethically supportable, and computationally tractable with existing tools. Its honest current form is an in-silico study that generates falsifiable predictions for subsequent empirical work—not a device proposal. That is not a diminished ambition. Given the distance between what is aspired to and what is currently possible, it is the contribution the field actually needs.

Note on sources. All references are peer-reviewed unless otherwise indicated. Bibliographic details should be verified against the primary sources prior to submission or citation in derivative work.

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